Vending Machine Project

Updated March 4, 2026

By Nick Phan

Designed to accept American Quarters

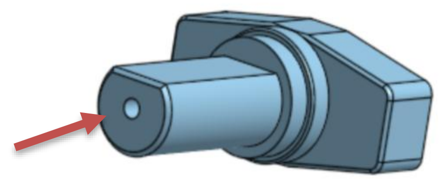
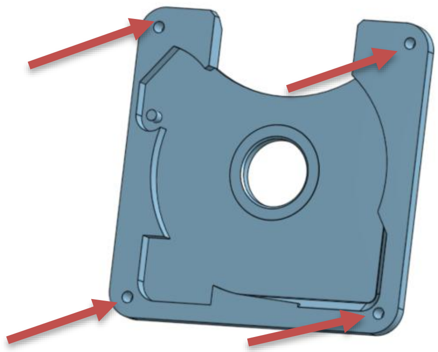
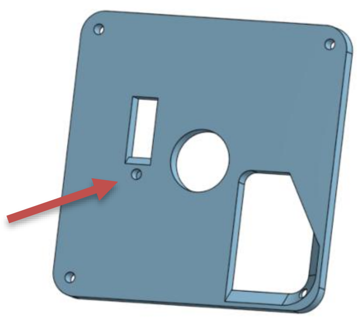
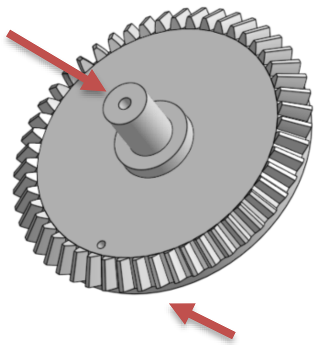
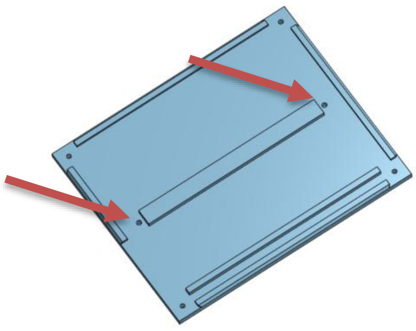
Required Materials:

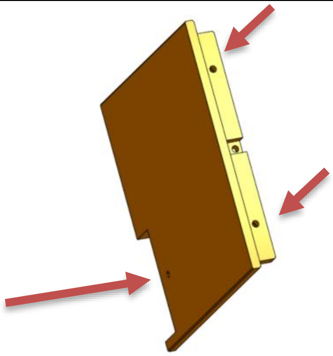
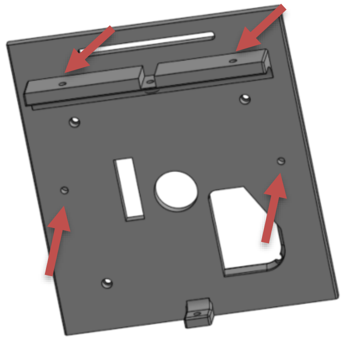
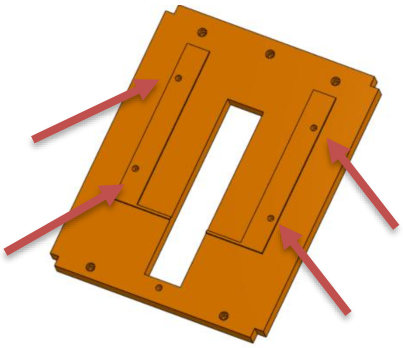
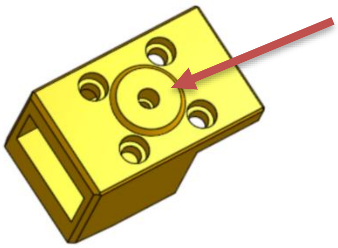
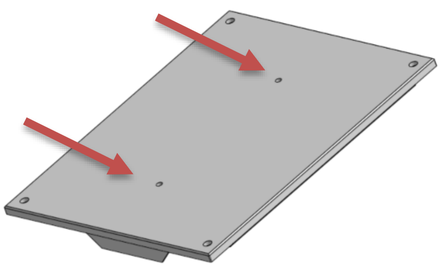
1 200mm MGN09 linear rail with MGN9H carriage block
https://www.amazon.com/dp/B0D54L45WM
24 m3x5x4 heat set inserts.
https://www.amazon.com/Threaded-Inserts-Soldering-Printed-Materials/dp/B0D7M3LJDL?sr=8-8
24 M3 x 10mm cap head screw
https://www.amazon.com/Fgruh-750PCS-Assortment-Washers-Assorted/dp/B0FGV5FCBN?sr=8-4
6 M3 x 6mm cap head screw
https://www.amazon.com/Fgruh-750PCS-Assortment-Washers-Assorted/dp/B0FGV5FCBN?sr=8-4
4 400mm 2020 extrusion with both ends tapped for M5 threads. (linked product does not have threads)
https://www.amazon.com/Aluminum-Extrusion-European-Standard-Anodized/dp/B0B7DXYFZ5?sr=8-11
8 m5 x 12mm cap head screws
https://www.amazon.com/Socket-Screws-Grade-Thread-Quantity/dp/B07NTYF2FZ?sr=8-4
(optional) 1/16”(1.5mm) clear acrylic, 160mm x 220 for top front display.
https://www.amazon.com/Polycarbonate-Resistant-Plexiglass-Robotics-Industrial/dp/B07VWLCJN6?sr=8-5
*__2__ 6mm x 200mm x 400mm panels (for side panels)
*__1__ 6mm x 160mm x 230mm panel (for back top panel)
**__1__ 75mm x 150mm piece of acrylic
**__1__ 5/8" Long Cam Lock for Cabinets
https://www.amazon.com/dp/B0DDJS11ZD
**glue that bonds to plastic.
Grease lubricant
https://www.amazon.com/Super-Lube-21030-Synthetic-Grease/dp/B000XBH9HI?sr=8-4
3" x 4.5" cardboard sleeves (this system holds 100 cards at a time)
https://www.amazon.com/gp/product/B01MR92HHX?smid=A244BBLLP9BF04
some sort of weights, I use **20** 2" washers
https://www.amazon.com/dp/B00L1IXL3S
1 1/4" x 2" or m5 x 50mm bolt (to be used as a system lock when it runs out of cards to dispense)
*If you are not 3d printing the side panels
**for the optional quick access top back panel

Tools:

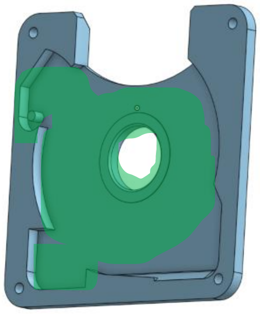
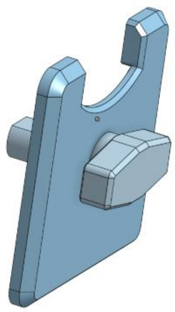
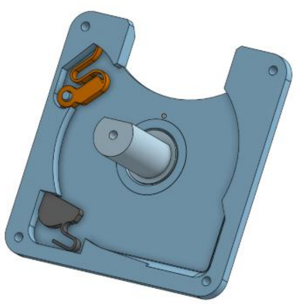
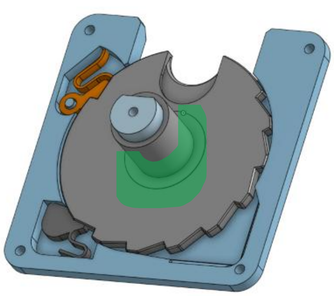
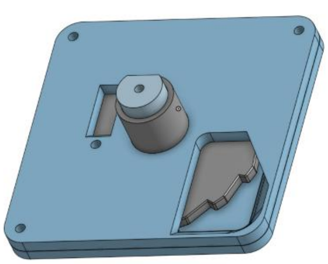
Solder Iron/m3 heat set insert tool
2mm hex wrench/screw driver
4mm hex wrench/screw driver
3d printer capable of printing 240mm x 240mm x 180mm
M5 thread tap (if tapping your own threads into the 2020 extrusions.

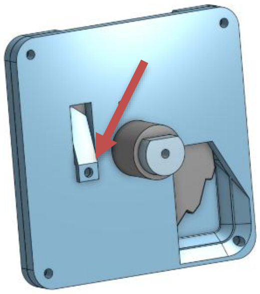
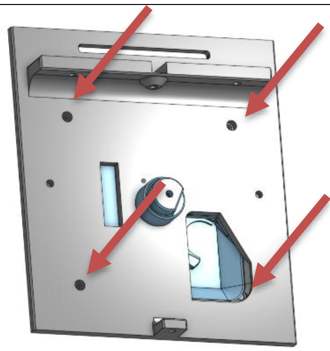
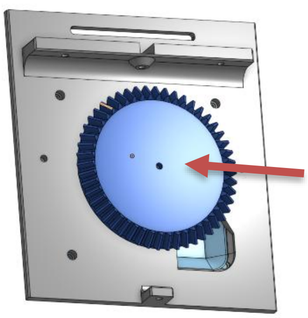
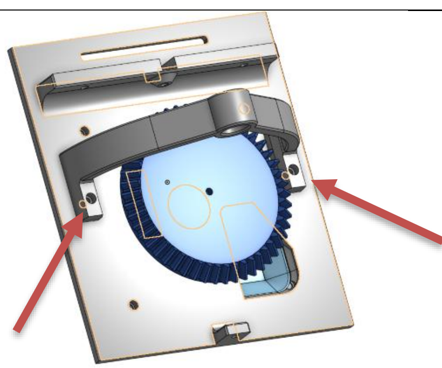
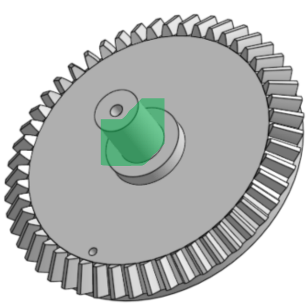
Prep: Heat Set inserts

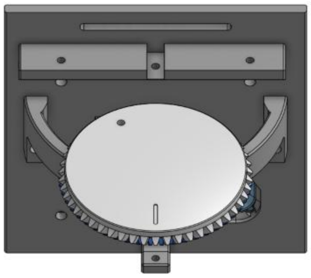
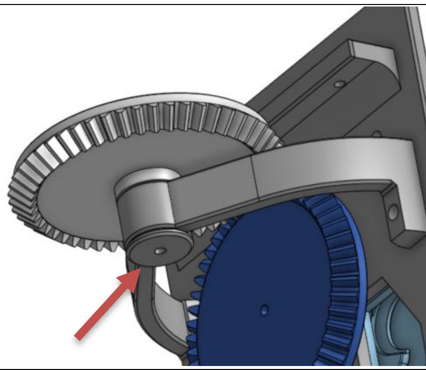
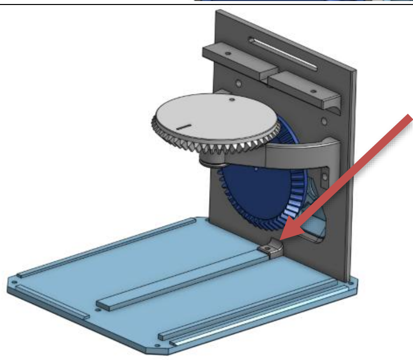
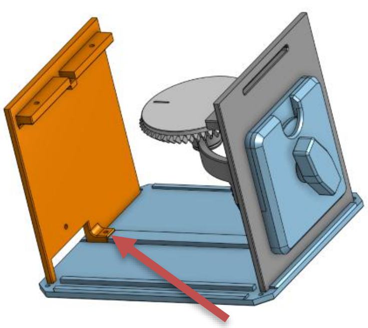
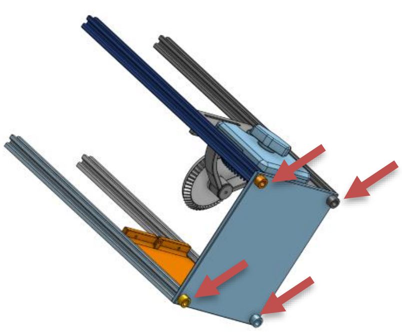
	**Step 1:** Insert 1 m3 heat set insert into the **crank.**
	**Step 2:** Insert 4 m3 heat set inserts into the **front plate.**
	**Step 3:** Insert 1 m3 heat set insert into the **back plate.**
	**Step 4:** Insert 1 m3 heat set insert into the axle of the **output gear.** Insert 1 m3 heat set insert into the hole on the flat end of the **output gear.**
	**Step 5:** Insert 2 m3 heat set inserts into the **floor panel.**

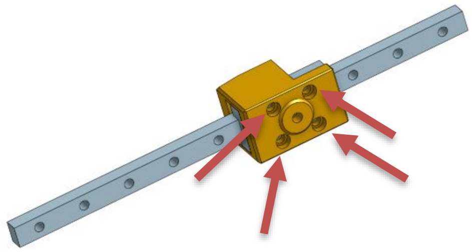
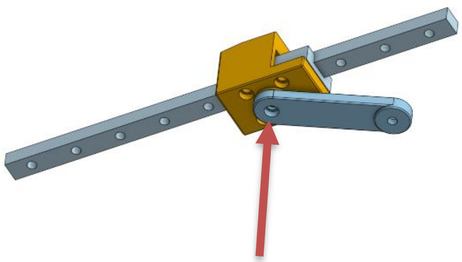
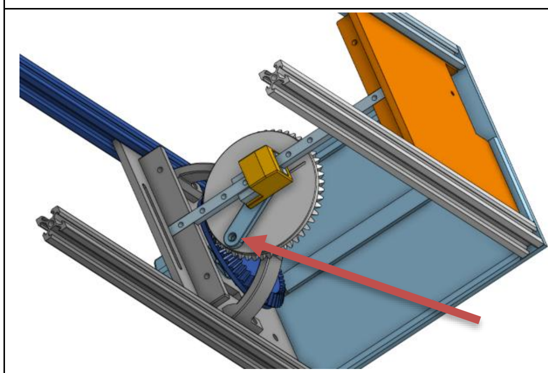
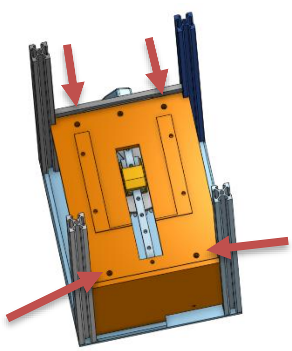
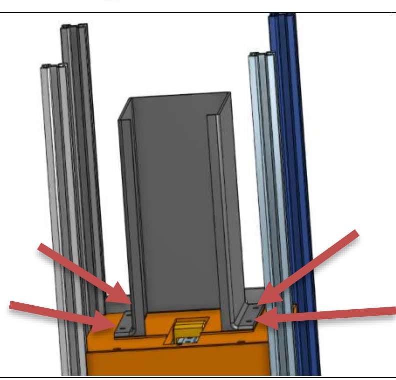
	**Step 6:** Insert 3 m3 heat set inserts into the **back bottom panel.**
	**Step 7:** Insert 4 m3 heat set inserts into the **front bottom panel.**
	**Step 8:** Insert 4 m3 heat set inserts into the **card shelf.**
	**Step 9:** Insert 1 m3 heat set insert into the **card pusher**.
	**Step 10(optional):** If you want to attach the handle to the top, insert 2 m3 heat set inserts into the **ceiling panel.**

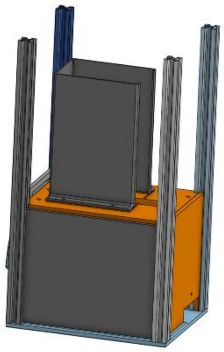
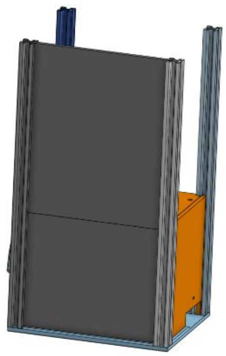
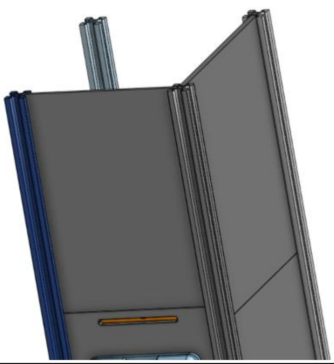
Assembly

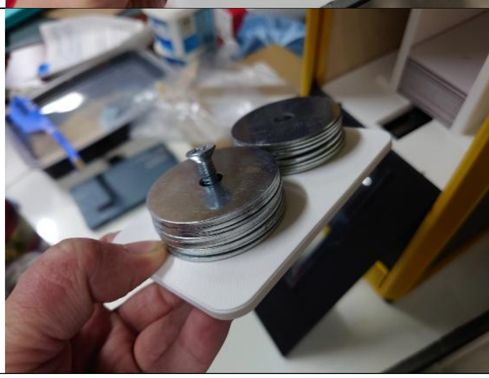
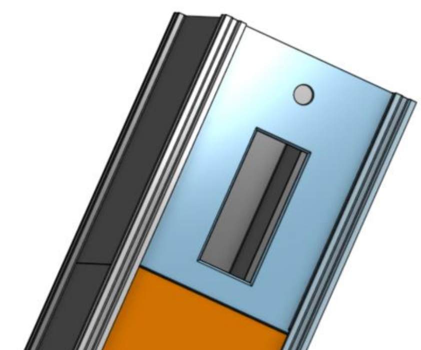
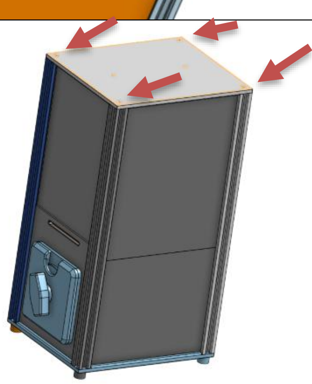
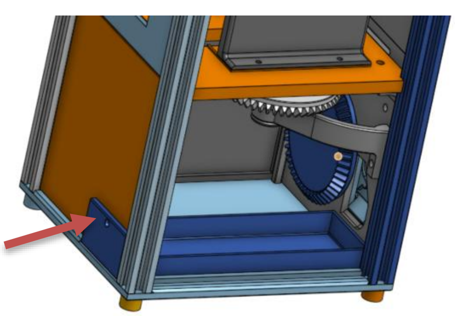
	**Step 1:** Lubricate the inside of the **front plate.**
	**Step 2**: Insert the **crank**.
	**Step 3**: Insert the **coin diameter checker** and **ratchet**.
	**Step 4:** Insert your desired **ratchet gear**. Lubricate the axle of the **ratchet gear**.
	**Step 5:** Sandwich the coin mechanism assembly with the **back plate**

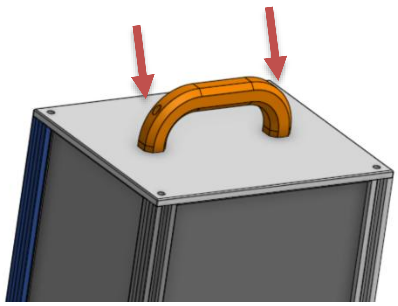
	**Step 6:** (optional for 25c and 50c) Use 1 m3 x 10 cap head screw to attach the **coin thickness checker** to the **back plate**. If you are using 75c or 4-quarter setups, I would recommend attaching the **coin thickness checker**. This stops the machine if there are not two quarters in the slot.
	**Step 7:** Use 4 m3 x 10 cap head screws to attach the coin mechanism to the **front bottom panel**.
	**Step 8:** Use 1 m3 x 10 cap head screw to attach the **input gear** to the back of the **crank**.
	**Step 9:** Use 2 m3 x 10 cap head screws to attach the **output gear gantry** to the **front bottom panel**.
	**Step 10:** Lubricate the **output gear** axle**.**

	**Step 11:** Insert the **output gear** into the **output gear gantry** with the notch facing perpendicular to the **front bottom panel**.
	**Step 12:** Use 1 m3 x 10 cap head screw to attach the **output gear washer** to the bottom of the **output gear**.
	**Step 13:** Use 1 m3 x 10 cap head screw to attach the **bottom front panel** to the **floor panel.**
	**Step 14:** Use 1 m3 x 10 cap head screw to attach the **bottom back panel** to the **floor panel**.
	**Step 15:** Use 4 m5 x 12 cap head screws to attach the feet and 2020 extrusions to the floor panel.

	**Step 16:** Use 4 m3 x 6 cap head screws to attach the **card pusher** to the MGN09H carriage.
	**Step 17:** Use 1 m3 x 10 cap head screw to attach the flat end of the **piston** to the **card pusher**.
	**Step 18:** Set the MGN09 linear rail in the notch of the **bottom front panel** and **bottom back panel**. Use 1 m3 x 10 cap head screw to attach the **piston** to the **output gear**.
	**Step 19:** Place the **card shelf** on the ledges provided by the **bottom front panel** and the **bottom back panel**. Make sure the orientation is as shown. The **card pusher** should sit in the window in the middle, and the open end of the U should face the front of the machine. Use 4 m3 x 10 cap head screws to attach the **card shelf** to the **bottom front panel** and the **bottom back panel**.
	**Step 20:** Attach the **Card Stack holder** to the **Card shelf** with 4 m3 x 10 cap head screws.

	**Step 21:** Slide the **bottom side panel** down the 2020 extrusion channel as shown. Repeat for both sides.
	**Step 22:** Slide the **top side panel** down the 2020 extrusion channel as shown. Repeat for both sides.
	**Step 23:** Slide the **top front panel** down the 2020 extrusion channel as shown.
	**Step 24:** Insert your 1.5mm(1/16”) 160 x 220mm acrylic panel over top the **top front panel** to protect the front art.
	**Step 25(optional):** Glue your 75mm x 150mm (3” x 6”) piece of acrylic in window slot of **the top back panel.** Attach your 5/8” cabinet lock into the top hole of the **top back panel.**

	**Step 26:** Load your cards into the **card stack holder.** I find it more reliable to load the card holders with the fold facing the front.
	**Step 27:** Place the **card stack weight** on top of the stack of cards. I use 20 2inch washers glued to the **card stack weight** to give it weight. I also have a 2inch bolt sitting in the hole of the **card stack weight**. When there are no more cards left in the **card stack holder**, it will drop down and block the mechanism from turning, preventing people from wasting their money.
	**Step 28:** Slide the **top back panel** into the extrusion channels as shown. If you are using the 3d printed hinge one, then it will fall down until the top is attached.
	**Step 29:** Use 4 m5 x 12mm cap head screws to attach the **ceiling panel** to the 4 2020 extrusions.
	**Step 30:** Use 1 m3 x 10mm cap head screw to fix the **coin collector** in place.

	**Step 31(optional):** Use 2 m3 x 10mm cap head screws to attach the **travel handle** to the **ceiling panel.**

Designer's notes:

I designed this machine because I could not find any sticker vending machines for sale around me. It isn't designed to be a machine left out in the wild to make money. It was more designed as a fun interactive object I could keep with me when I am at craft fairs.

I decided to make a vending machine that dispenses cards with a turning mechanism, commonly found on gumball dispensers, because I figured the sliding mechanism traditionally found on these machines would not work with such a lightweight object. I imagined you would push the whole machine around the table if you put the traditional coin slider mechanism.

I prioritized these mechanisms to use real coins rather than proprietary tokens because I think the satisfying clunk-plink noise the coin makes when dropping down is an important part of using the machine. Thus, I never spent any time designing a unique token.

I incorporated the cabinet lock as a latching mechanism for quickly restocking the machine. It wasn't really designed as a security feature. I figured security features would be rather moot because the whole mechanism weighs next to nothing, and a thief could pick up the machine and run away with little to no effort.

If the mechanism is jamming or gets locked before it can dispense a card, it is usually one of 3 problems. There are no more cards in the card stack holder, and the bolt from the card stack weight is blocking the mechanism. Second, if the bottom of the card stack holder has some elephant's foot from the printing, it can sometimes catch a card and prevent it from dispensing. You can use a knife or deburring tool to remove any elephant's foot from the card stack holder to stop that from happening. The third possibility is that there is a coin that failed to exit the coin mechanism and did not fall into the coin collector. Some wiggling of the crank or a slap on the side of the machine can help the coin fall, and the machine will become operational again.

If the crank is able to turn 360 degrees when there are no coins inserted, it is probably because the coin diameter checker is stuck in the up position. This happens if the machine is not set on a level surface, does not have enough lubrication to freely fall into place, or when the coin mechanism is fastened too tightly to the bottom front panel, causing the coin diameter checker to bind.

Vending machine as a business card dispenser

Using a 3d printed token as currency for the machine

Turning the mechanism

Once the crank is turned 360 degrees, a card will dispense

Pull the card out of the machine. The machine cannot dispense another card until this card is removed.

Revealing the card inside the card holder